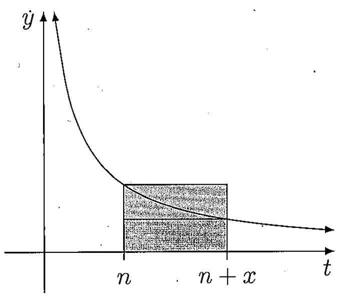
Inequalities WS 8

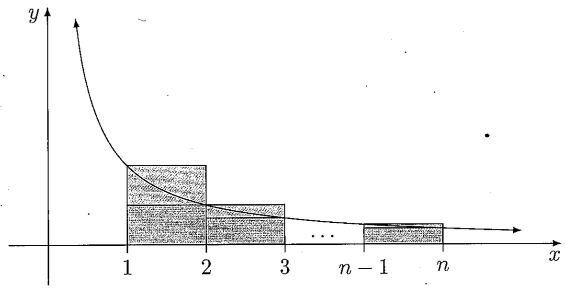
1. Let $a, b, c > 0$
   a. Show that $a^3 + b^3 \geq abc\left(\frac{a}{c} + \frac{b}{c}\right)$
   b. Hence, show that $\frac{a^3+b^3+c^3}{3} \geq abc$
   c. Deduce that $\frac{a^3}{b} + \frac{b^3}{c} + \frac{c^3}{a} \geq ab + bc + ac$
2. Let $a, b, c \geq 0$ and $abc = 1$. Prove that $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}}$
3. Let $a, b, c > 0$. Prove that $a(a-b)(a-c) + b(b-a)(b-c) + c(c-a)(c-b) \geq 0$.
4. Define a sequence of angles $\theta_k \in \left[0, \frac{\pi}{2}\right]$ where $k = 1, 2, 3, \ldots, n$ such that $0 < \theta_1 + \theta_2 + \theta_3 + \cdots + \theta_n < \frac{\pi}{2}$

   Use mathematical induction to prove that $\tan(\theta_1 + \theta_2 + \theta_3 + \cdots + \theta_n) \geq \tan\theta_1 + \tan\theta_2 + \tan\theta_3 + \cdots + \tan\theta_n$
5. Prove the following inequalities
   a. $\forall x > 0, x \geq \ln(1+x)$
   b. $\forall x > 0, e^x > 1 + x + \frac{x^2}{2}$
   c. $\forall x > 0, \sin x > x - \frac{x^3}{6}$
   d. $\forall x > 0, x < \frac{e^x - e^{-x}}{2}$
   e. $\forall x \in \mathbb{R}, \cos x \geq 1 - \frac{x^2}{2}$
   f. $\forall x \in \left[0, \frac{\pi}{2}\right], x < \ln(\sec x + \tan x)$
   g. $\forall x \in [0,1], \sin^{-1} x \geq x$
   h. $\forall x > 0, \tan^{-1} x \geq x - x^3$
6. Let $f(x) = x^n e^{-x}$
   a. Show that there is a maximum turning point at $(n, n^n e^{-n})$
   b. Sketch the graph for $x \geq 0$
   c. Deduce that $\left(1 + \frac{1}{n}\right)^n < e$
7. The diagram below shows a section of the graph of $y = \frac{1}{t}$.



   Consider the region $t \in [n, n+x]$, where $x > n$.
   a. Prove that $\frac{x}{1+\frac{x}{n}} < n\ln\left(1 + \frac{x}{n}\right) < x$
   b. Hence, show that $\lim_{n\to\infty}\left(1 + \frac{x}{n}\right)^n = e^x$

8. The diagram below shows the graph of $y = \frac{1}{x}$. Upper and lower-bound rectangles of unit width are constructed over the domain $x \in [1, n]$.



Define the series $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$.

a. Show that $\frac{1}{n} + \ln n < H_n < 1 + \ln n$

b. Hence, find two integers which are lower and upper bounds of the following sum.

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2020}$$

c. Does the series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$ have a finite limit?

9. Let $0 \le t \le 1$

a. Prove that $\frac{1}{2} \le \frac{1}{1+t} \le 1$

b. Deduce that for $0 \le x \le 1$ $\quad \frac{x}{2} \le \ln(1+x) \le x$

10. Let $0 \le t \le 1$

a. Prove that $\frac{1}{2} \le \frac{1}{1+t^2} \le 1$

b. Deduce that for $0 \le x \le 1$ $\quad \frac{x}{2} \le \tan^{-1}(x) \le x$

11. Let $x > 0$ and $n \in \mathbb{Z}^+$

a. Prove that $1 - x \le \frac{1}{1+x} \le 1$.

b. Show that $1 - \frac{1}{2n} \le n \ln\left(1 + \frac{1}{n}\right) \le 1$.

c. Deduce that $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$

12. Define the integral for $n \ge 2$ $\quad I_n = \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^n}} dx$

a. Show that if $m > n$, then $I_m < I_n$

b. Hence, show that $\frac{1}{2} < I_n \le \frac{\pi}{6}$ for all $n \ge 2$

Answer:

8. a. proving; b. 7 and 9; c. no since H_n grows asymptotically with $\ln n$ according to (a), and $\ln n$ diverges as $n \to \infty$